CNN Architecture Comparison on CIFAR-10

Benchmarking nine architectures from LeNet to DenseNet on CIFAR-10 under a controlled training recipe to isolate the contribution of architecture design.

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Overview

CIFAR-10 is a standard 10-class image classification benchmark with 60,000 32x32 colour images. This project trains and compares nine architectures — a fully connected baseline, LeNet-5, VGG19, ResNet18, SENet18, ResNeXt, DenseNet, GoogLeNet, and a Dual Path Network — under identical conditions on an RTX 3080 GPU. All models share the same optimiser (SGD with momentum), cosine learning-rate schedule, and standard CIFAR augmentation so that differences in test accuracy can be attributed to architecture rather than training setup.

Goals

Provide a fair, controlled comparison of convolutional architecture families on CIFAR-10, and identify which design choices (depth, residual connections, dense connectivity, grouped convolutions) deliver the best accuracy per parameter within a fixed 20-epoch training budget.

Objectives

- Implement nine architectures (FC Net, LeNet-5, VGG19, ResNet18, SENet18, ResNeXt, DenseNet, GoogLeNet, DPN) adapted for CIFAR-10 in a shared cifar_models.py module.
- Train every model with an identical recipe — SGD with momentum, cosine LR schedule, random crop and horizontal flip augmentation applied on-device — for a fair comparison.
- Record test accuracy, parameter count, and training time for each model in results.json.
- Analyse the accuracy-per-parameter trade-off across architecture families and identify the key inductive biases driving performance gains.

Approach

Images are loaded as normalised GPU tensors with on-device augmentation to keep the data pipeline fast. Each architecture is trained for 20 epochs and evaluated on the held-out test set. Results are written incrementally to results.json so no progress is lost. A final comparison table and discussion summarise how each design choice (skip connections, dense reuse, squeeze-and-excitation attention, grouped convolutions) affects test accuracy and parameter efficiency.

Outcomes

- SENet18 achieved the highest test accuracy at 0.930 with 11.3M parameters.
- ResNet18 and DenseNet both reached 0.925 accuracy; DenseNet did so with only 1.00M parameters — the best accuracy-per-parameter of any model tested.
- The FC baseline plateaued at 0.499 and LeNet-5 at 0.707, confirming that convolutional inductive bias and depth are essential for CIFAR-10.
- All modern architectures (VGG19 through DPN) clustered between 0.895 and 0.930, showing that

residual and dense connectivity, not raw parameter count, drives accuracy in this regime.